Lathe Machine

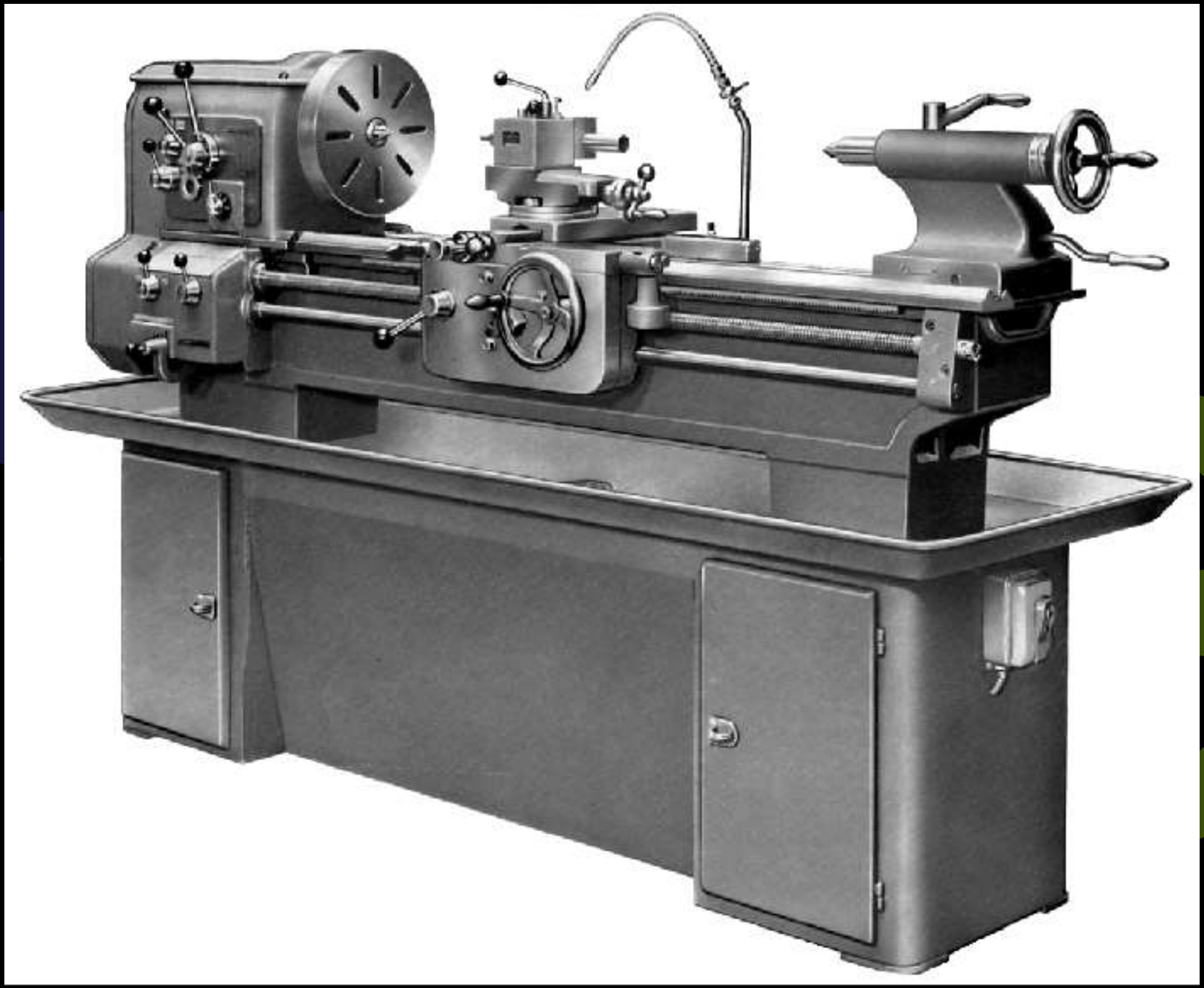
History

- Lathe forerunner of all machine tools
- First application was potter's wheel
  - Rotated clay and enabled it to be formed into cylindrical shape
- Very versatile
- Used for turning, tapering, form turning, screw cutting, facing, drilling, boring, spinning, grinding and polishing operations

Types of Lathes

- Engine lathe
  - Not production lathe, found in school shops, toolrooms, and job shops
  - Primarily for single piece or short runs
  - Manually operated

Engine Lathe

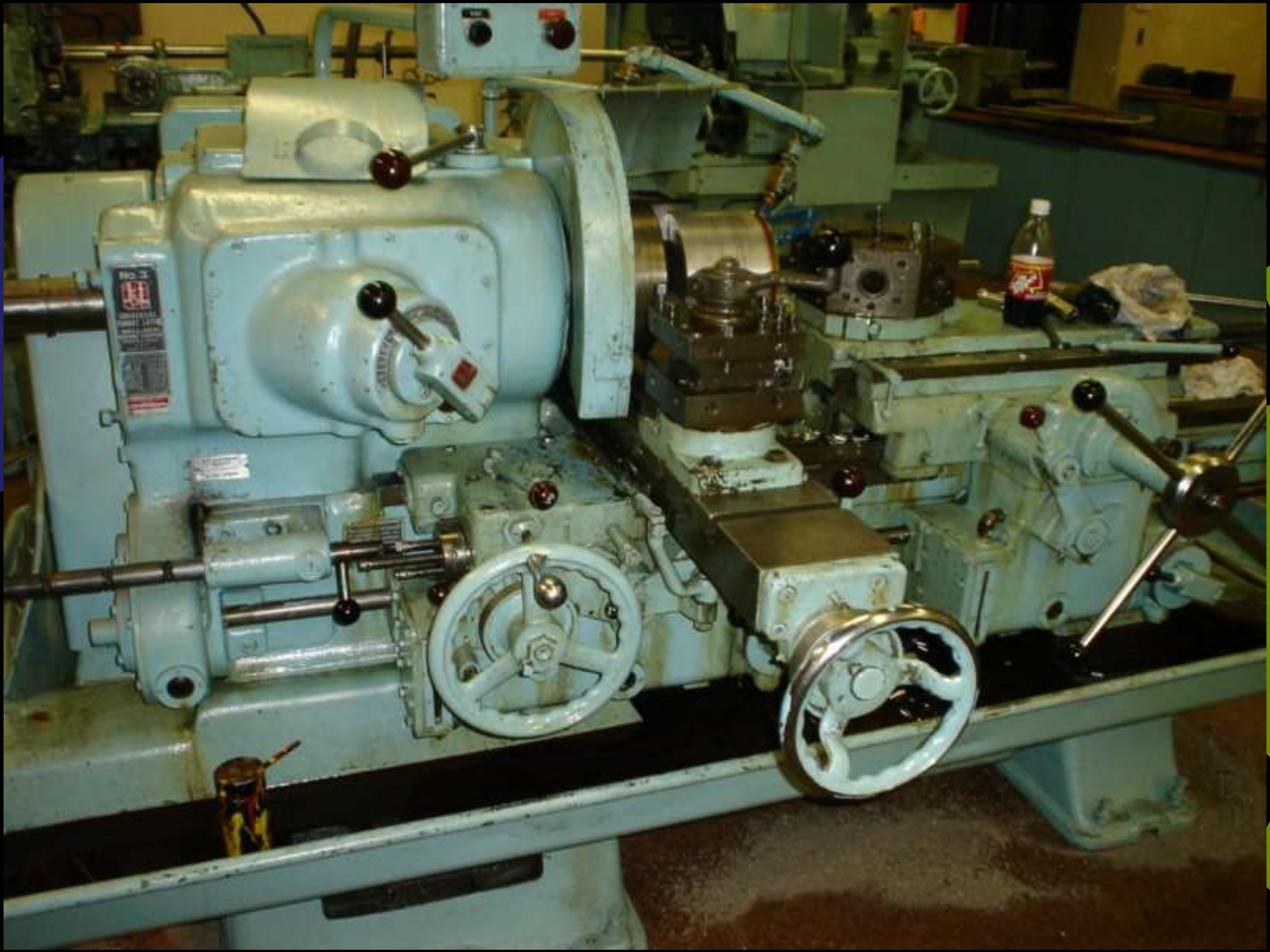


Special Types of Lathes

Turret lathe

- Used when many duplicate parts required
- Equipped with multisided toolpost (turret) to which several different cutting tools mounted
  - Employed in given sequence

Turret Lathe



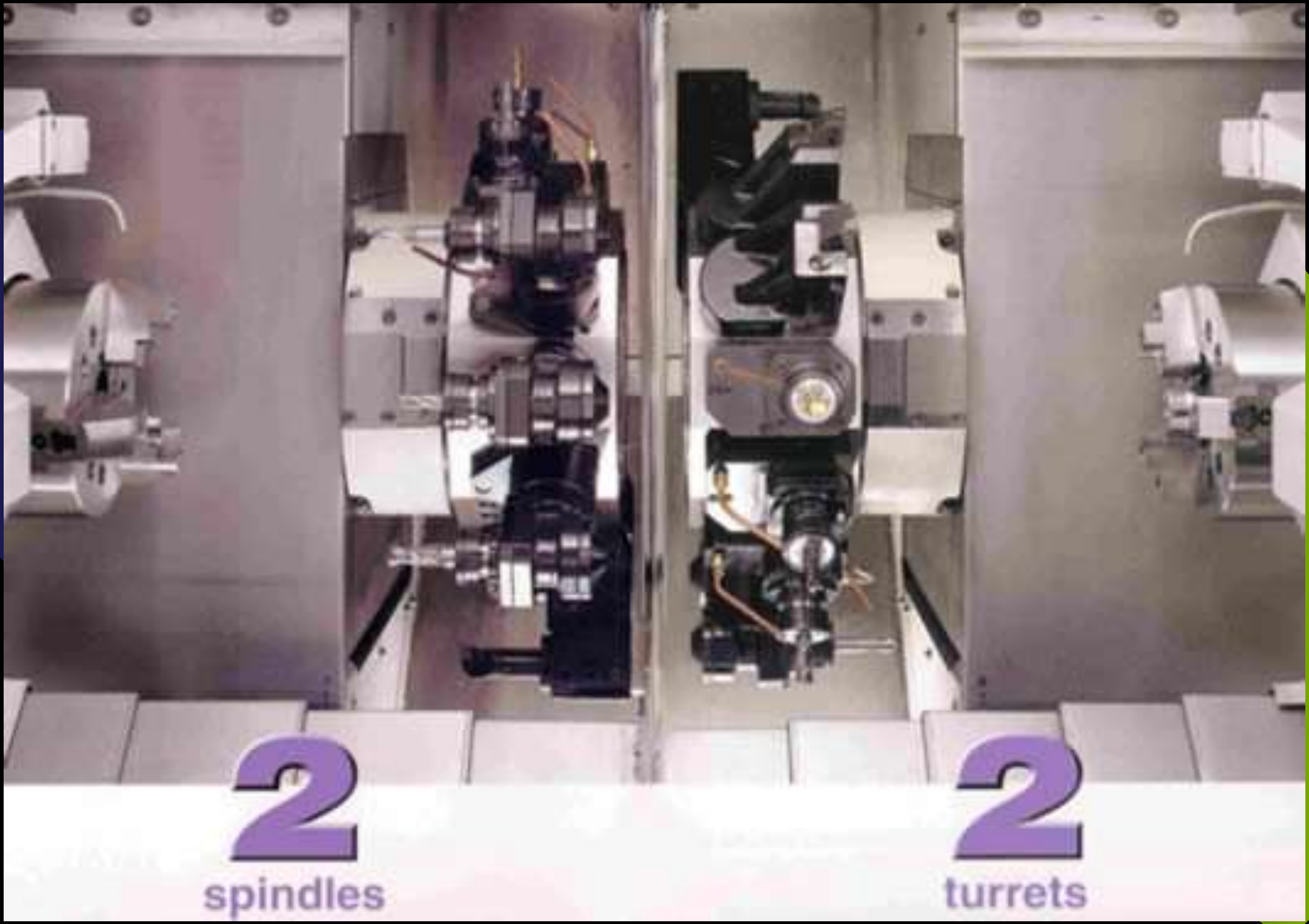
Special Types of Lathes

- Single- and multiple-spindle automatic lathes
  - Six or eight different operations may be performed on many parts at the same time
  - Will produce parts for as long as required
- Tracer lathes
  - Used where a few duplicate parts required
  - Hydraulically operated cross-slide controlled by stylus bearing against round or flat template

Special Types of Lathes

- Conventional/programmable lathe
  - Operated as standard lathe or programmable lathe to automatically repeat machining operations
  - 2-axis (DRO) so can see exact location of cutting tool and workpiece in X and Z axes
- Computerized numerically controlled lathes
  - Cutting-tool movements controlled by computer-controlled program to perform sequence of operations automatically

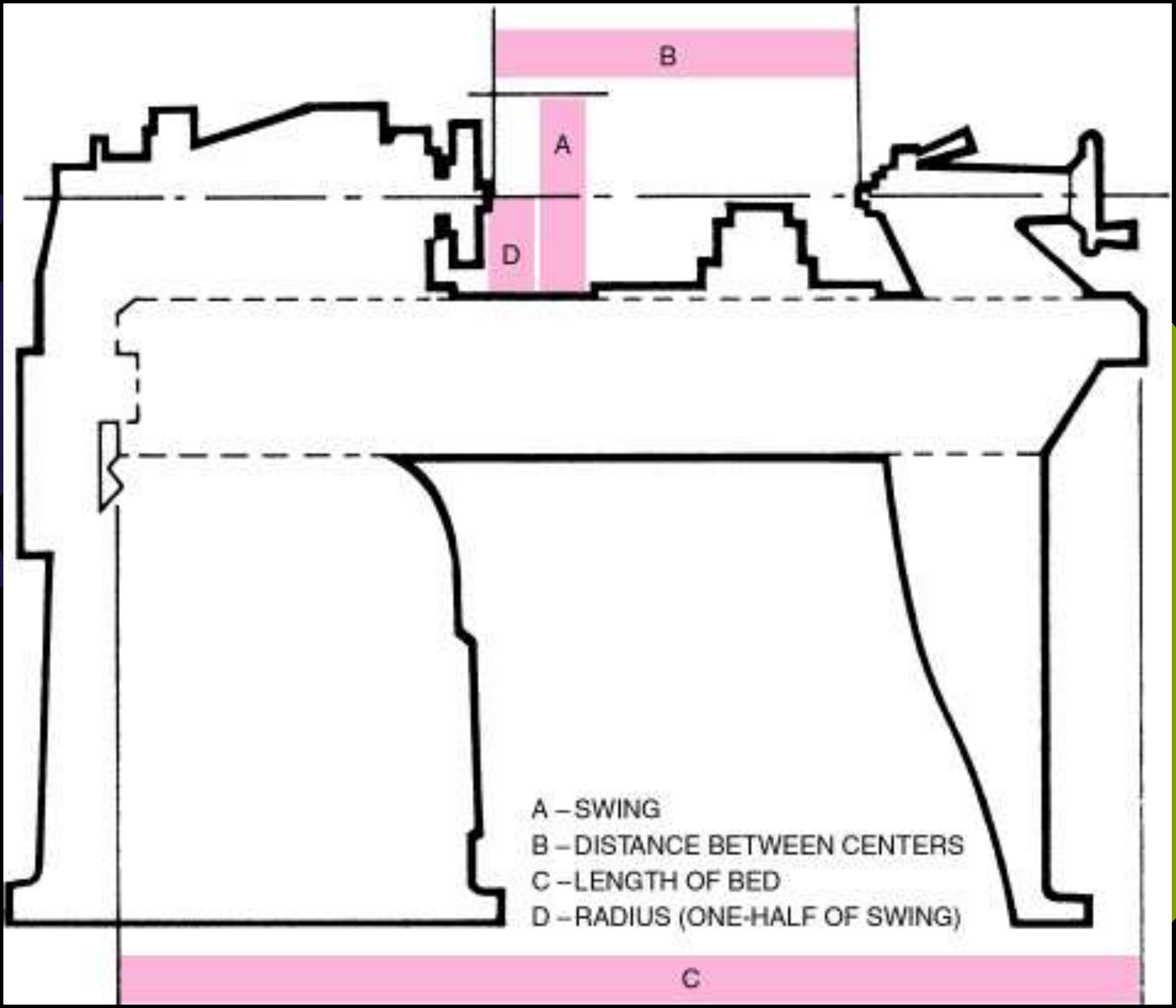
CNC Lathe



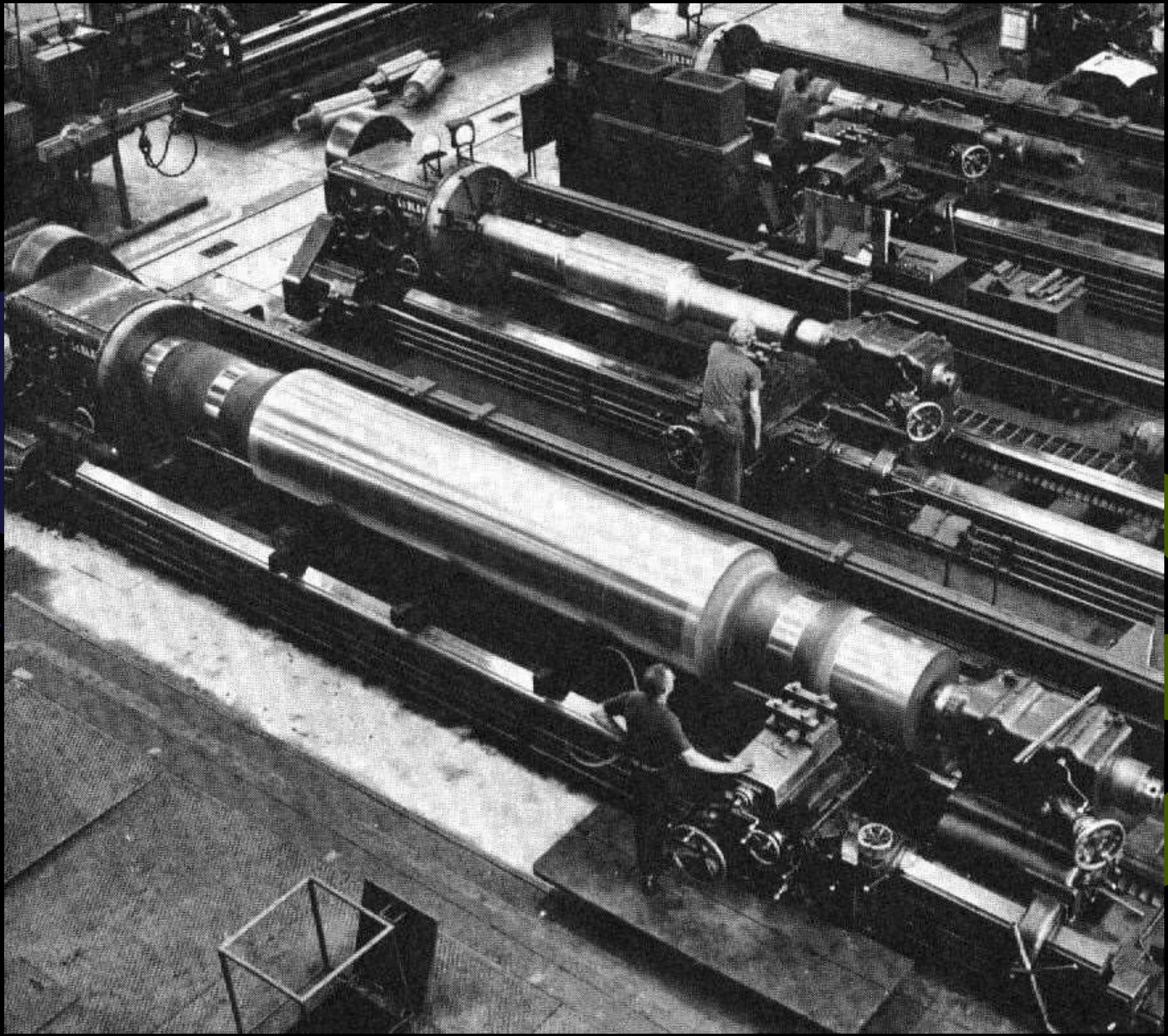
Lathe Size and Capacity

- Designated by largest work diameter that can be swung over lathe ways and generally the maximum distance between centers
- Manufactured in wide range of sizes
  - Most common: 9- to 30- in. swing with capacity of 16 in. to 12 feet between centers
  - Typical lathe: 13 in. swing, 6 ft long bed, 36 in.
  - Average metric lathe: 230-330 mm swing and bed length of 500 – 3000 mm

Lathe Size


B
A
D
A – SWING
B – DISTANCE BETWEEN CENTERS
C – LENGTH OF BED
D – RADIUS (ONE-HALF OF SWING)
C

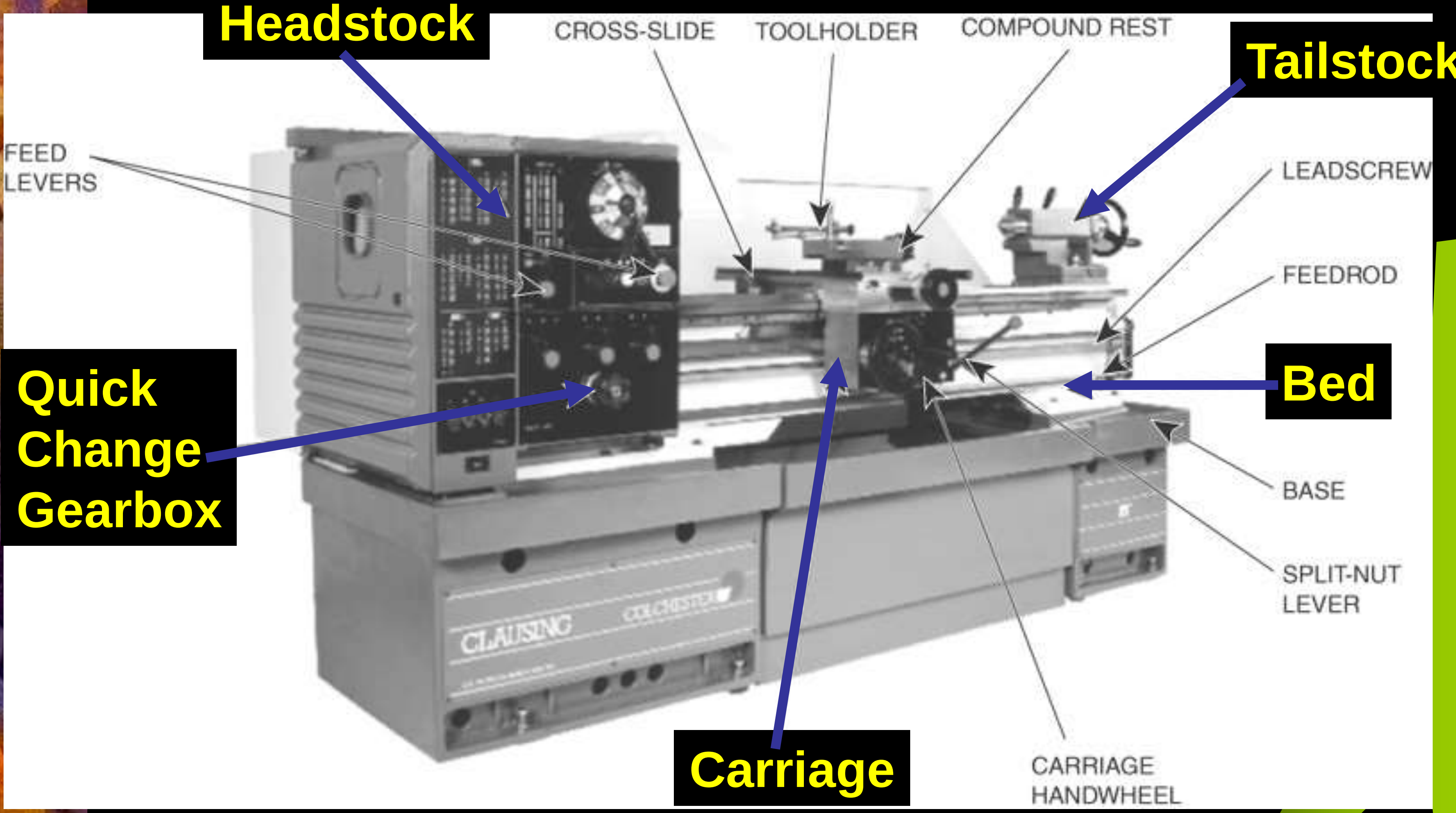
Lathe Size



Lathe Size

Lathe Size

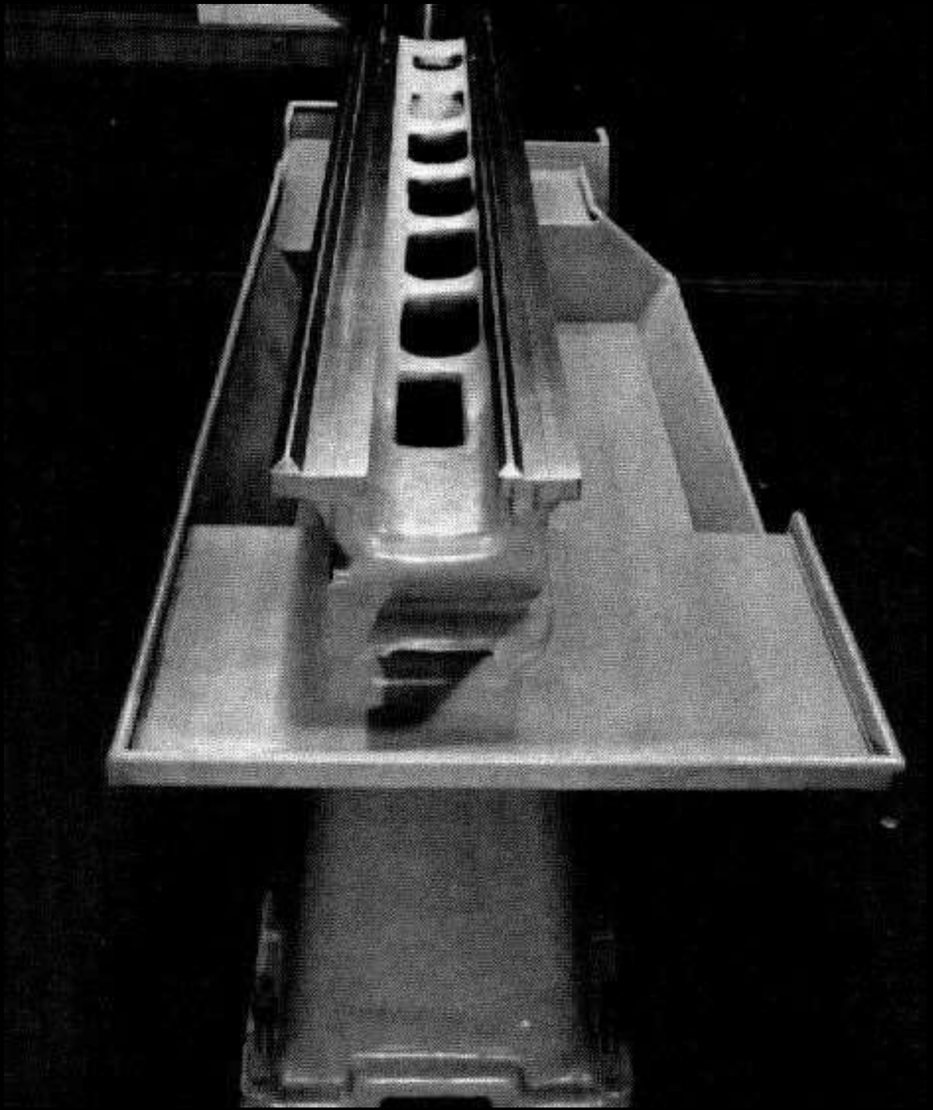
Parts of the Lathe



Lathe Bed

- Heavy, rugged casting
- Made to support working parts of lathe
- On top section are machined ways
    - Guide and align major parts of lathe

Lathe Bed



Headstock

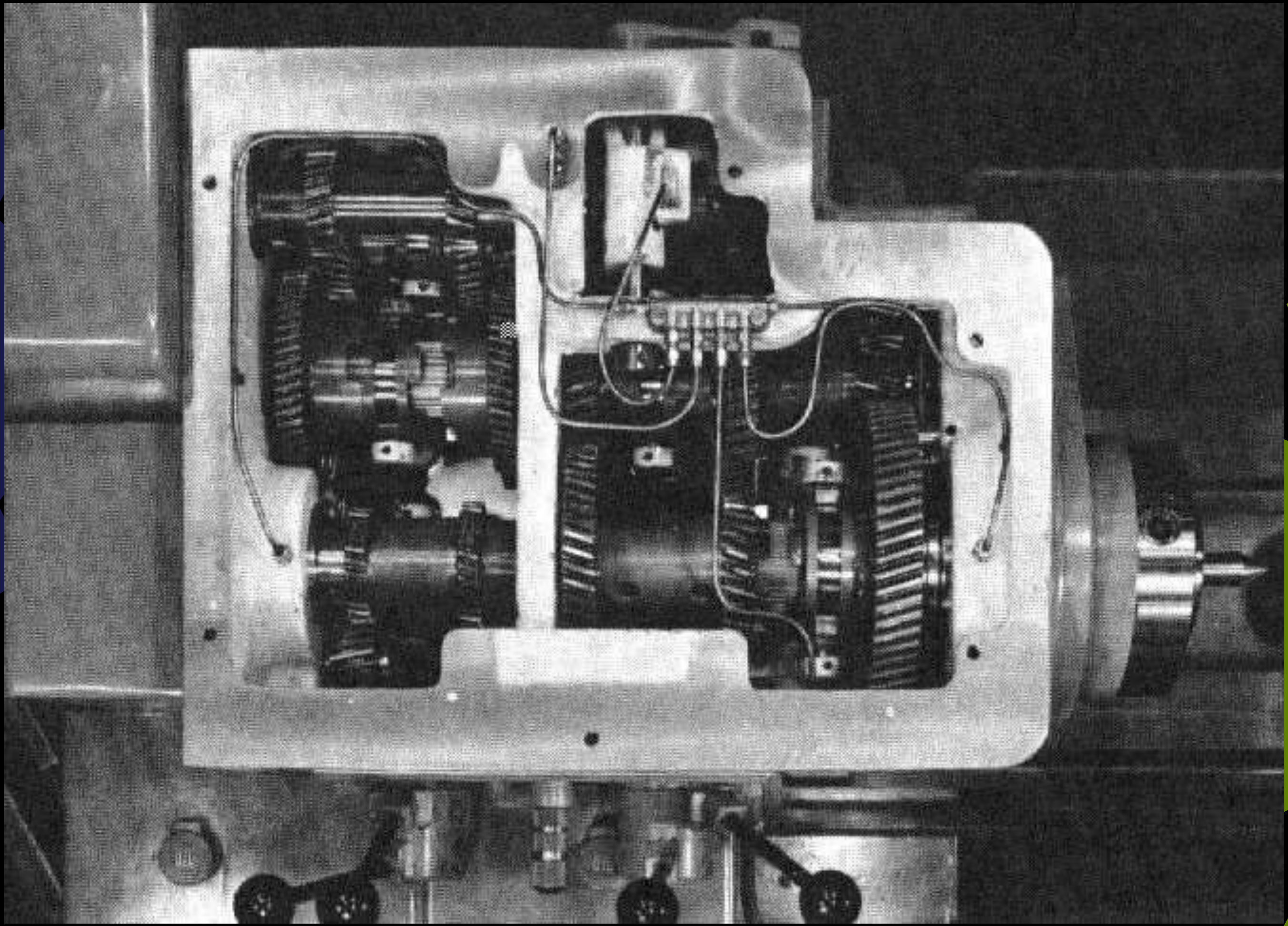
- Clamped on left-hand end of bed
- Headstock spindle
  - Hollow cylindrical shaft supported by bearings
    - Provides drive through gears to work-holding devices
  - Live center, faceplate, or chuck fitted to spindle nose to hold and drive work
- Driven by stepped pulley or transmission gears
- Feed reverse lever
  - Reverses rotation of feed rod and lead screw

Headstock

Quick-Change Gearbox

- Contains number of different-size gears
- Provides feed rod and lead-screw with various speeds for turning and thread-cutting operations
  - Feed rod advances carriage when automatic feed lever engaged
  - Lead screw advances the carriage for thread-cutting operations when split-nut lever engaged

Quick-Change Gearbox

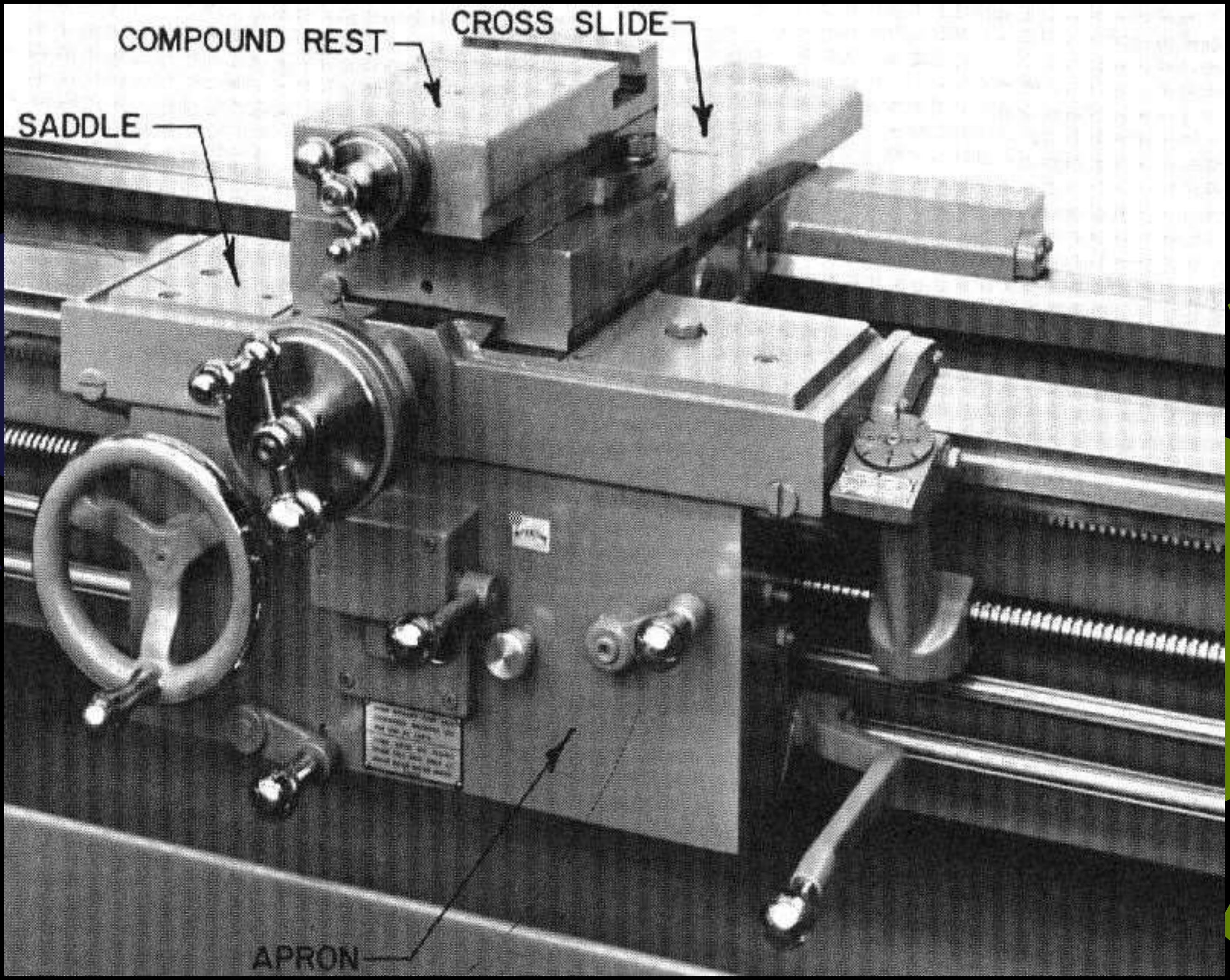


Top View

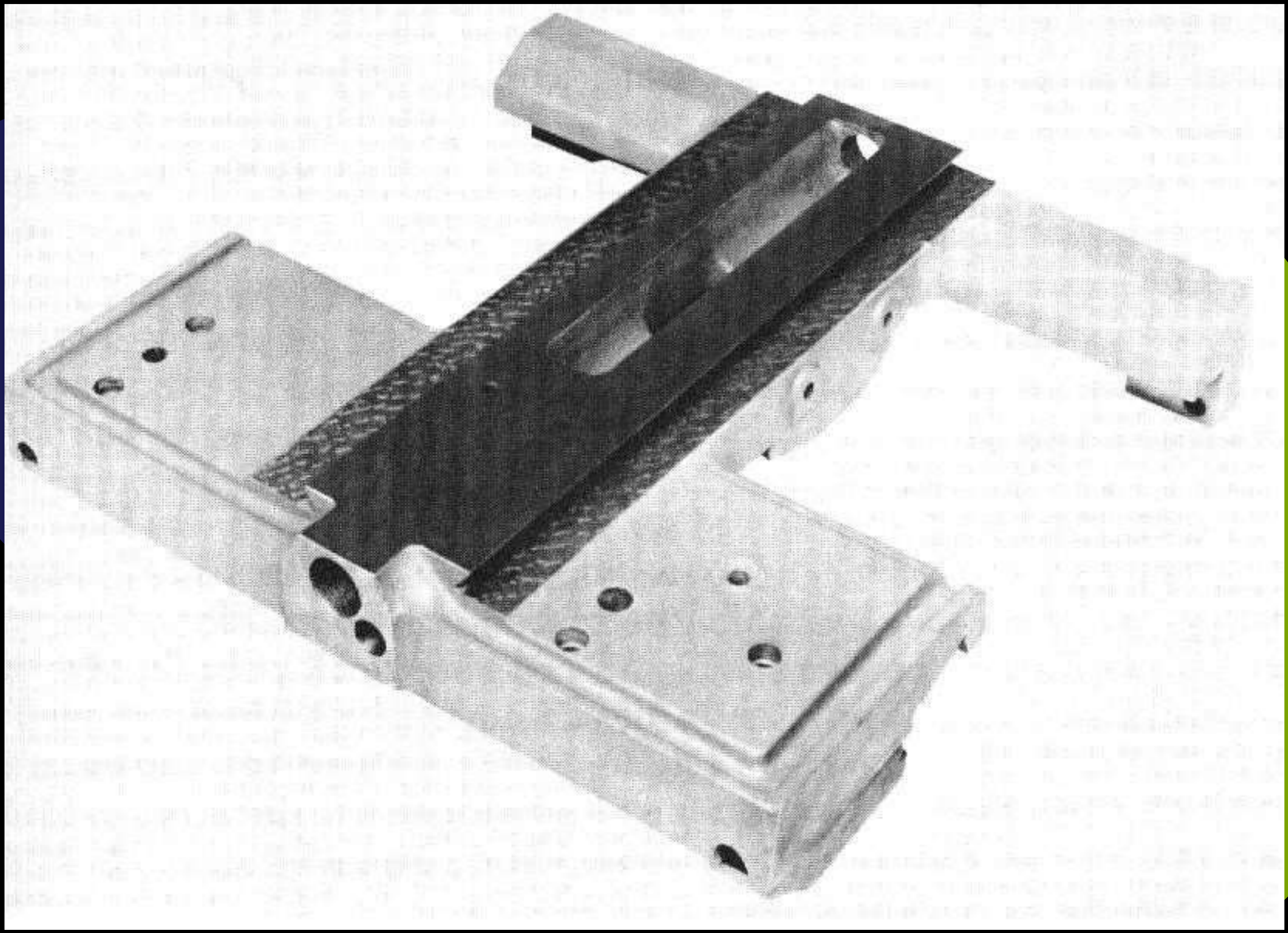
Carriage

- Used to move cutting tool along lathe bed
- Consists of three main parts
  - Saddle
    - H-shaped casting mounted on top of lathe ways, provides means of mounting cross-slide and apron
  - Cross-slide
  - Apron

Carriage



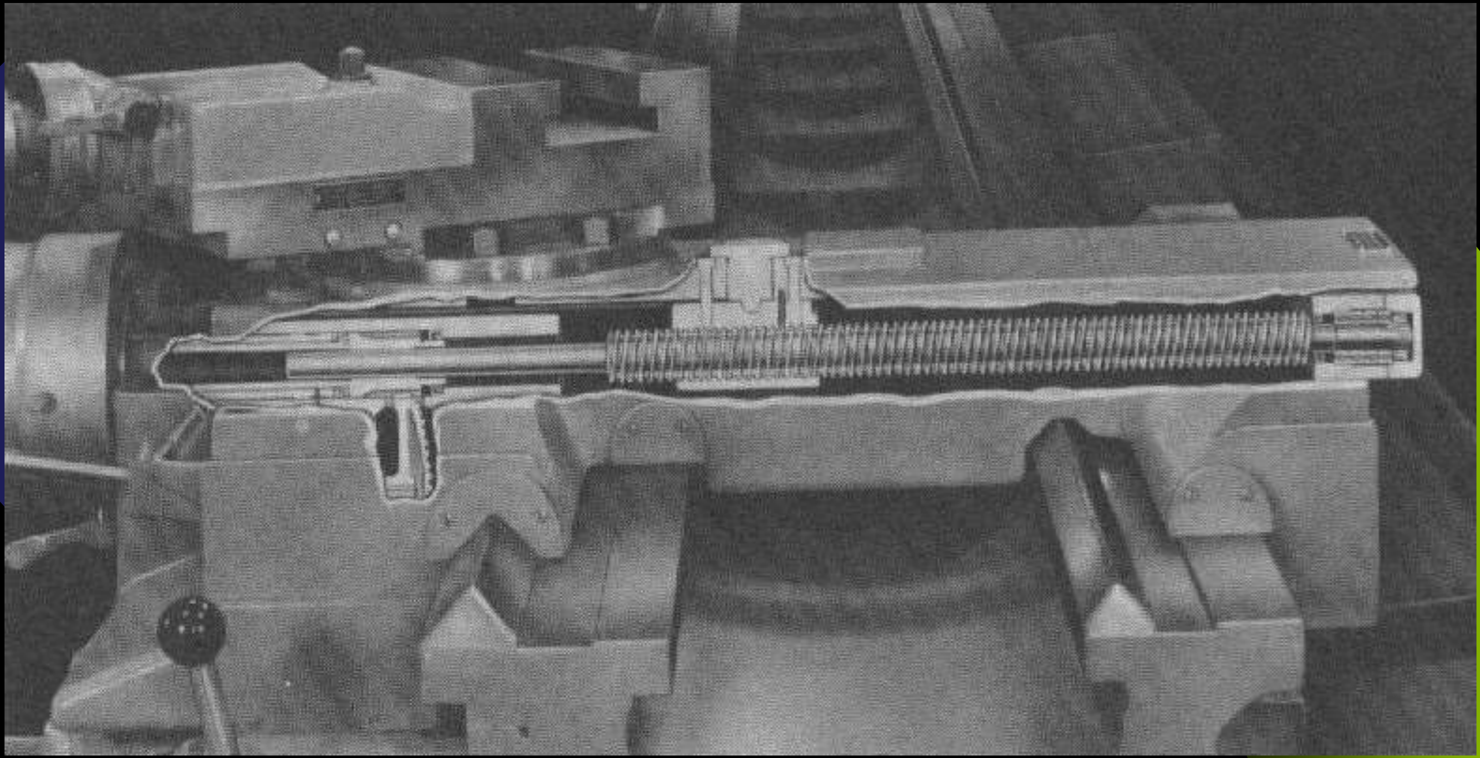
Carriage



Cross-slide

- Mounted on top of saddle
- Provides manual or automatic cross movement for cutting tool
- Compound rest (fitted on top of cross-slide)
  - Used to support cutting tool
  - Swiveled to any angle for taper-turning
  - Has graduated collar that ensure accurate cutting-tool settings (.001 in.) (also cross-slide)

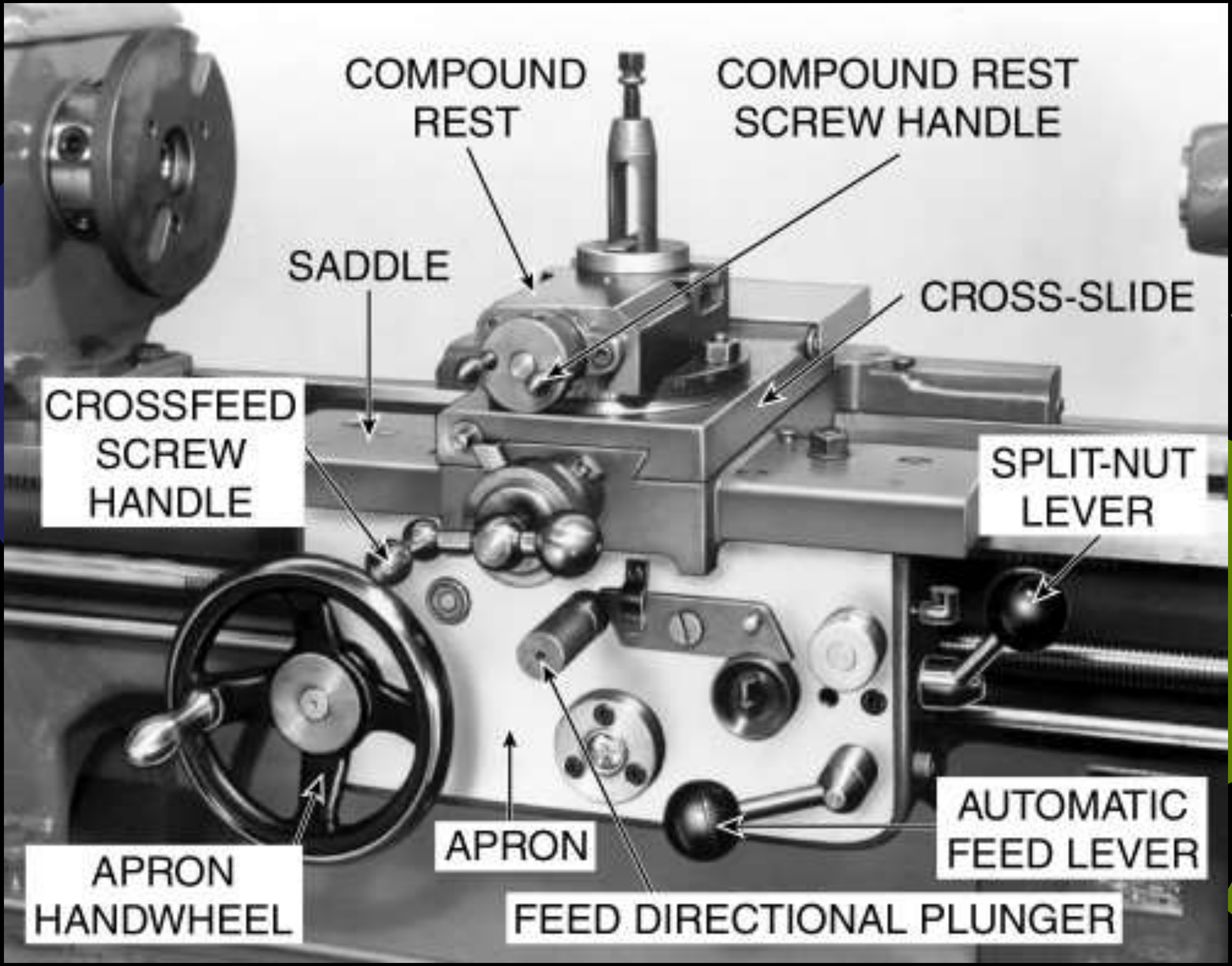
Cross-slide



Apron

- Fastened to saddle
- Houses gears and mechanism required to move carriage or cross-slide automatically
- Locking-off lever inside apron prevents engaging split-nut lever and automatic feed lever at same time
- Apron handwheel turned manually to move carriage along lathe bed

Apron


COMPOUND REST
COMPOUND REST SCREW HANDLE
SADDLE
CROSS-SLIDE
CROSSFEED SCREW HANDLE
SPLIT-NUT LEVER
APRON
AUTOMATIC FEED LEVER
APRON HANDWHEEL
FEED DIRECTIONAL PLUNGER

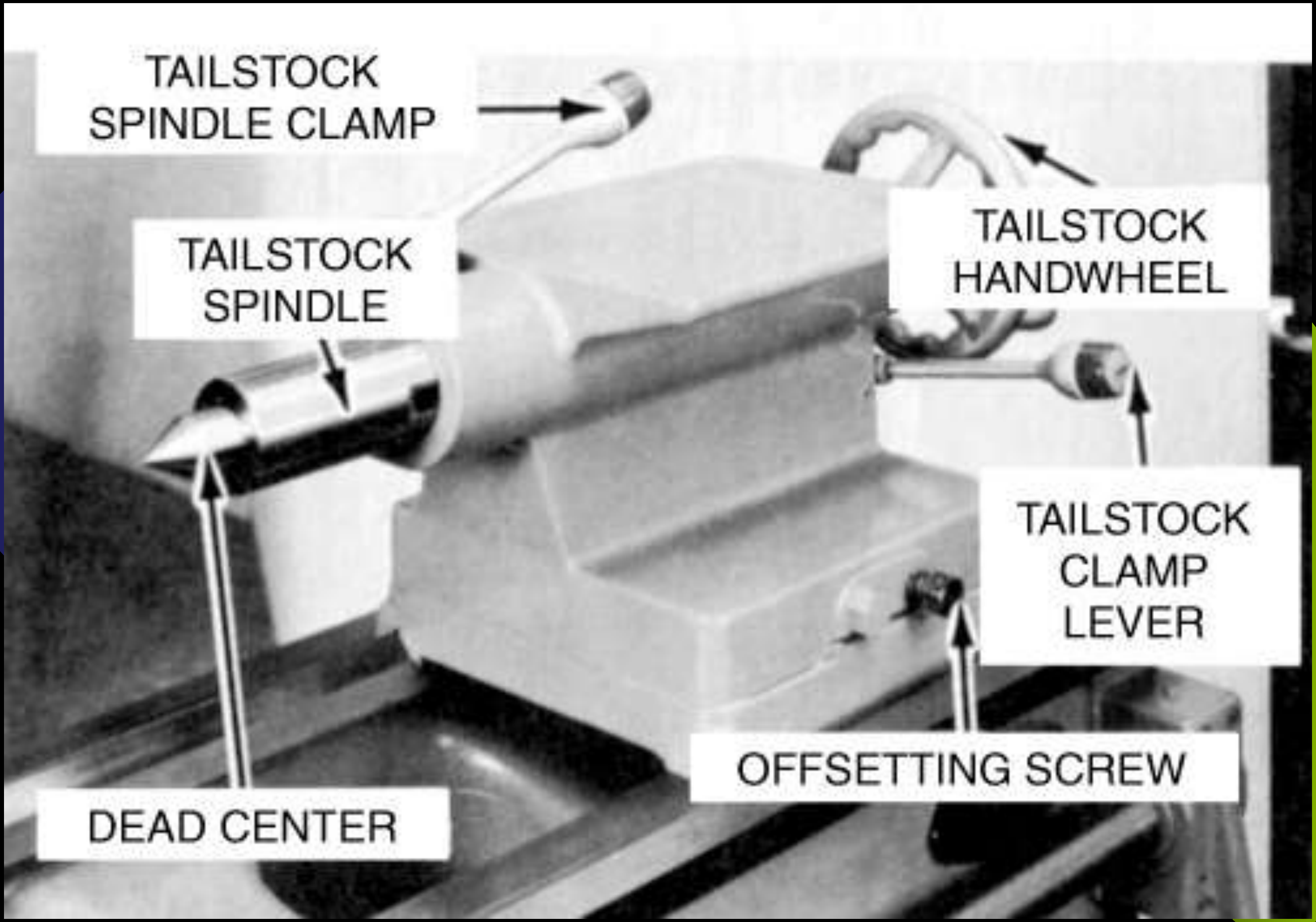
Automatic Feed Lever

- Engages clutch that provides automatic feed to carriage
- Feed-change lever can be set for longitudinal feed or for crossfeed
  - In neutral position, permits split-nut lever to be engaged for thread cutting
  - Carriage moved automatically when split-nut lever engaged

Tailstock

- Upper and lower tailstock castings
- Adjusted for taper or parallel turning by two screws set in base
- Tailstock clamp locks tailstock in any position along bed of lathe
- Tailstock spindle has internal taper to receive dead center
  - Provides support for right-hand end of work

Tailstock



TAILSTOCK SPINDLE CLAMP
TAILSTOCK SPINDLE
TAILSTOCK HANDWHEEL
TAILSTOCK CLAMP LEVER
OFFSETTING SCREW
DEAD CENTER

Setting Speeds on a Lathe

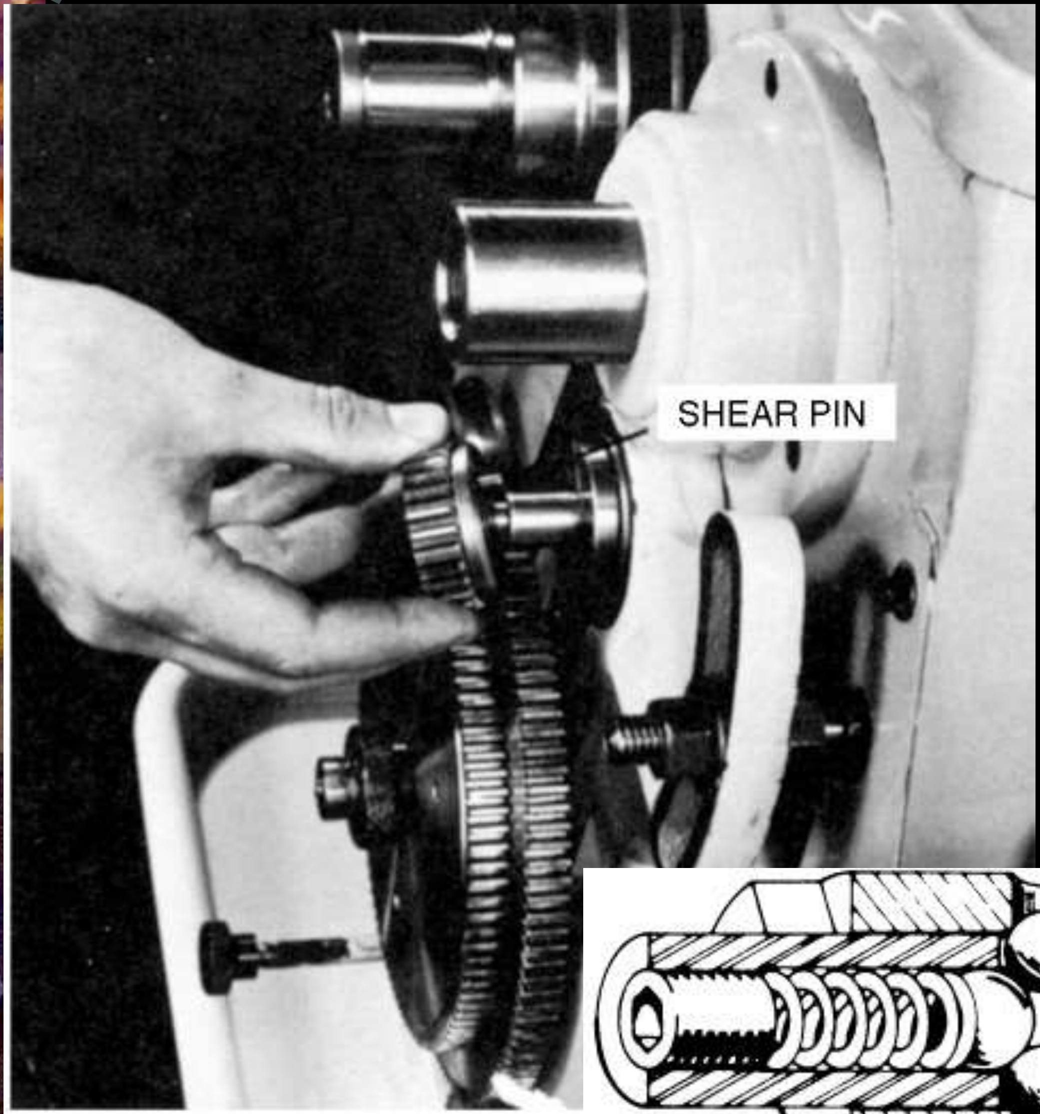
- Speeds measured in revolutions per minute
  - Changed by stepped pulleys or gear levers
- Belt-driven lathe
  - Various speeds obtained by changing flat belt and back gear drive
- Geared-head lathe
  - Speeds changed by moving speed levers into proper positions according to r/min chart fastened to headstock

Feed of an Engine Lathe

- Distance carriage will travel in one revolution of spindle
- Depends on speed of feed rod or lead screw
  - Controlled by change gears in quick-change gearbox
    - Obtains drive from headstock spindle through end gear train
  - Chart mounted on front of quick-change gearbox indicates various feeds

Shear Pins and Slip Clutches

- Prevents damage to feed mechanism from overload or sudden torque
- Shear pins
  - Made of brass
  - Found on feed rod, lead screw, and end gear train
- Spring-loaded slip clutches
  - Found only on feed rods
  - When feed mechanism overloaded, shear pin will break or slip clutch will slip causing feed to stop



Shear pin in end gear train prevents damage to the gears in case of an overload

Spring-ball clutch will slip when too much strain is applied to feed rod